CS/IS LRB

Reg. No. | | | | | | | | | | |

[17BS4MA01]

# FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

## LINEAR ALGBERA

**Time:3 Hours** **Maximum Marks:70**

**Instructions:**

1. Answer any **FIVE** full questions. Choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed.

## PART-A

**1. a.** For what values of k the equations $x+y+z=1; 2x+y+4z=k; 4x+y+10z=k^2$ have a solution and solve them completely in each case. **07**

**b.** Solve following system equations using LU decomposition method: **07**

$$2x_1-2x_2-2x_3=-4;$$
$$-2x_2+2x_3=-2;$$
$$-x_1+5x_2+2x_3=6;$$

**2. a.** Let $R^+$ be the set of all positive real numbers. Define vector addition as $u+v=uv$ $\forall u,v\in R^+$ and scalar multiplication by $k.u=u^k, \forall u\in R^+, k\in R$. Show that $R^+$ is vector space over the field of real numbers. **07**

**b.** Define subspace of vector space. Which of the following are subspaces: **07**

**i.** $W=\left\{\begin{bmatrix} x \\ y \end{bmatrix}, x+y=1\right\}$ in $R^2$.

**ii.** $W=\left\{\begin{bmatrix} x \\ y \\ z \end{bmatrix}, x^2+y^2+z^2\le 1\right\}$ in $R^3$.

**iii.** $W=\left\{\begin{bmatrix} x \\ y \\ z \end{bmatrix}, x+y+z=0\right\}$ in $R^3$.

**3. a.** Find the basis and dimension of the row space of the matrix A and column space of the matrix by taking transpose of A **07**

$$A=\begin{bmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 2 & -6 & 9 & -1 & 8 & 2 \\ 2 & -6 & 9 & 1 & 9 & 7 \\ -1 & 3 & -4 & 2 & -5 & -4 \end{bmatrix}.$$

**b.** Determine whether $(1,1,1,1)$, $(1,2,3,2)$, $(2,5,6,4)$ and $(2,6,8,6)$ form a basis of $R^4$. If not find the dimension of the subspace they span. **07**

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4. a. Define Linear transformation. Find the linear transformation $T: R^2 \to R^3$ such that $T(-1, 0) = (-1, 0, 2)$ and $T(2, 1) = (1, 2, 1)$. **07**

b. Let $W$ be a subspace of $R^5$ spanned by $u = (1, 2, 3, -1, 2)$ and $v = (2, 4, 7, 2, -1)$. Find a basis of $W$ orthogonal to $u$ and $v$. **07**

## PART-B

5. a. Define orthogonal projection and also find the projection of the vector $v = (1, -2, 3, 4)$ along $W = span\{w_1 = (1,1,0,3), w_2 = (1,2,-1,2)\}$ in $R^4$. **6**

b. Solve the following system of equations $AX = b$ by the method of least squares, where **8**

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 5 & 3 \\ 2 & 0 & 1 \\ 3 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} -1 \\ 2 \\ 0 \\ 1 \\ -2 \end{bmatrix}.$$

6. a. Using Gram-Schmidt process find an orthogonal basis and hence an orthonormal basis of the subspace $W$ spanned by the following vectors $v_1 = (1,1,1,1)), v_2 = (1,2,4,5)$ and $v_3 = (1,-3,-4,-2)$ of $R^4$. **6**

b. Find a modal matrix $P$ which diagonalize the matrix **8**

$$A = \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}.$$

7. a. Solve the following Linear programming problem by graphically

Minimize $z = 200x + 500y$

subjected to : **6**

$$x + 2y \geq 10,$$

$$3x + 4y \leq 24,$$

$$x \geq 0, y \geq 0.$$

b. A candy manufacturer has 130 pounds of chocolate-covered cherries and 170 pounds of chocolate-covered mints in stock. He decides to sell them in the form of two different mixtures. One mixture will contain half cherries and half mints by weight and will sell for \$2.00 per pound. The other mixture will contain one-third cherries and two-thirds mints by weight and will sell for \$1.25 per pound. How many pounds of each mixture should the candy manufacturer prepare in order to maximize his sales revenue? **8**

8 a. Write the characteristics of canonical and standard forms in simplex method. 4

b. Find the optimal solution for the following linear programming problem by using the simplex method, 10

Maximize; $Z = 6x_1 + 10x_2 + 4x_3$

Subject to :

$$x_1 + x_2 + x_3 \leq 1000$$

$$x_1 + x_2 \leq 500$$

$$x_1 + 2x_2 \leq 700$$

$$x_1, x_2, x_3 \geq 0.$$